Yohan Abeysinghe

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EDUCATION

University of Maryland, USA **August 2025 – Present**
Ph.D - Electrical and Computer Engineering *Advisor: Prof. Dinesh Manocha*

- Developing deep learning frameworks for reasoning over complex audio-visual scenes, with emphasis on temporal consistency, scene structure, and scalable visual perception.
- Researching vision-language-action models and flow-based generative policies for robot navigation, path planning, and scene-aware autonomous behavior.

University of Moratuwa, Sri Lanka **Jan 2020 – July 2024**
B.Sc (Hons) - Electronics and Telecommunication Engineering *3.80 (First Class Honours)*

Maliyadeva College, Kurunegala, Sri Lanka **Grad: Dec 2018**
GCE Advanced Level (Mathematics, Physics, Chemistry) *3As / z-score of 2.78 (Island 24)*

PUBLICATIONS

- Gershom Seneviratne, **Yohan Abeysinghe**, Jianyu An, Vaibhav Shende, Rahul Kumar, Dinesh Manocha. "VEGA: Learning Navigation VLAs from In-the-Wild Egocentric Video with Geometric Trajectory Supervision." *arXiv preprint arXiv:2606.18426*, 2026. [ArXiv] [Project Page]
- **Yohan Abeysinghe**, Muhammad Akhtar Munir, Sanoojan Baliah, Ron Sarafian, Fahad Shahbaz Khan, Yinon Rudich, Salman Khan. "SynCast: Synergistic Neural Forecasting of Air Pollution with Stochastic Sampling." *Nature NPJ.* [ArXiv]
- Sanjoy Chowdhury, Mohamed Elmoghany, **Yohan Abeysinghe**, Junjie Fei, Sayan Nag, Salman Khan, Mohamed Elhoseiny, Dinesh Manocha. "MAGNET: A Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over Multi-Video Haystacks." In *Advances in Neural Information Processing Systems (NeurIPS)* 2025. [Project Page]
- Sanoojan Baliah, **Yohan Abeysinghe**, Rusiru Thushara, Khan Muhammad, Abhinav Dhall, Karthik Nandakumar, Muhammad Haris Khan. "VFace: A Training-Free Approach for Diffusion-Based Video Face Swapping." In *Winter Conference on Applications of Computer Vision (WACV) 2026.*

RESEARCH EXPERIENCE

Mohamed bin Zayed University of Artificial Intelligence, UAE **July 2024 – July 2025**
Research Associate *Advisor: Dr. Salman Khan*

- Developed and evaluated deep learning models for spatiotemporal prediction and vision-language reasoning, with emphasis on large-scale data pipelines, model evaluation, and interpretable scene-level outputs.

University of Sydney, Australia **Jan 2023 – Sept 2023**
Research Assistant - Internship *Advisor: Dr. Anusha Vithana*

- Implemented ML-based structure prediction pipelines and compared intermediate model outputs against physics-based simulations to evaluate learned representations and real-world dynamical behavior.

AWARDS

Bronze Medal: International Mathematics Olympiad (Brazil)	2017
Honourable Mention: International Mathematics Competition for University Students (Bulgaria)	2022
Bronze Medal: Asia Pacific Mathematics Olympiad (Mexico)	2018
Gold Medal: Sri Lankan Physics Olympiad	2019
International Physics Olympiad: (Israel)	2019
Honourable Mention: Asian Physics Olympiad (Australia)	2019
World Rank 163: IEEEXtream 15.0	2023
Honourable Mention: International Mathematics Olympiad	2014, 2015, 2016
Silver Medal: World Youth Mathematics Inter-Cities Competition (Thailand)	2016
Bronze Medal: World Youth Mathematics Inter-Cities Competition (Korea)	2014

Silver Medal: International Mathematics Competition (Bulgaria) 2013
Honourable Mention: Asia Pacific Mathematics Olympiad (Kazakhstan) 2015
Merit: International Mathematics Competition (Taipei, Taiwan) 2012
Bronze Medal: International Mathematics and Science Olympiad (Naga City, Philippines) 2011
Bronze Medal: Asia Inter-cities Teenagers Mathematics Olympiad (Kathmandu, Nepal) 2011

OTHER SELECTED PROJECTS

Vision-based Object Recognition and Task Solving Robot **Feb 2022 – Aug 2022**
- Built a vision-guided robot navigation pipeline in Webots using camera inputs and OpenCV/C++, connecting visual perception outputs to autonomous task execution.

AHEAD Intelligent Camera Project **Dec 2022 – Mar 2023**
- Implemented real-time YOLO-based object detection and face recognition on an embedded Axis P1341 camera, optimizing vision inference for resource-constrained deployment.

Implementing Iconic Papers In Machine Vision **Dec 2023 - Mar 2024**
- Implemented YOLO object detection from scratch to understand CNN-based detection architectures, localization losses, and end-to-end visual perception pipelines. **GitHub**
- Implemented Restormer to study efficient transformer architectures for dense visual prediction, focusing on long-range pixel interactions with reduced computational complexity. **GitHub**
- Implemented CoCa-style contrastive captioning architecture to explore multimodal representation learning with image-text supervision. **GitHub**

Head Pose Estimation Using Facial Landmark and Vision Graph Neural Networks **2024**
- Designed a vision-based head pose estimation model using facial landmark geometry, regional feature grouping, and lightweight Deep Sets for structured spatial reasoning.

Attention TCN for Motor Imagery Classification and Ear-EEG Acquisition Device **2024**
- Classifying Ear-EEG P300 signals from visual cues to identify targets vs non-targets. Using Scalograms with Grab-cad method to identify the heat maps and important frequency ranges in P300 classification process.
- Developed attention-based temporal convolution models for sequence classification, emphasizing temporal feature extraction and interpretable model behavior
- Developing an Ear-EEG acquisition device with a generic earpiece to make a new dataset.

Protein Folding: Free Energy Landscape with ML **2023**
- Implementing RoseTTAFold protein folding algorithm and taking the intermediate structures that a protein goes through while the algorithm is predicting the final structure.
- Using GROMACS Molecular Dynamics simulations to identify the folding pathways of a protein. Checking the similarities between the intermediate stages from two methods to identify how much of real world physics is learned by folding algorithms.

Project Eyeguard **Mar 2021 - Feb 2022**
- Developing a device to minimize the eye defects that can arise from long-term screen usage, by using cameras to track the eye and the blinking rates, and sending feedback to the users to take actions accordingly.

SKILLS

- Python
- PyTorch
- TensorFlow
- MATLAB
- ROS

RELEVANT COURSEWORK

Computer Vision: EN4553 Machine Vision (A+), EN4583 Advances in Machine Vision (A+), EN4573 Pattern Recognition and Machine Intelligence(A+)
Signal Processing: ENEE620 Random Processes in Communication and Control (A-), ENEE627 Information Theory (A-), ENEE630 Advanced Digital Signal Processing (A)
Mathematics: MA2023 Calculus (A+), MA2033 Linear Algebra (A), MA4013 Linear Models and Multivariate Statistics (A), MA2013 Differential Equations (A)

LEADERSHIP/EXTRACURRICULARS

President: Mathematics Society, University of Moratuwa 2021
Student Representative: Department of Electronic and Telecommunication, University of Moratuwa 2022
Committee Member: Electronics Club, University of Moratuwa 2023
Sergeant at Arms: Gavel Club, University of Moratuwa 2022
People Management Team: AIESEC in Colombo South 2021
Prefect: Maliyadeva College 2018

REFERENCES

Prof. Salman Khan
Associate Professor
Computer Vision Department
Mohamed bin Zayed University of Artificial Intelligence
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Prof. Dinesh Manocha
Distinguished University Professor
Department of Computer Science
University of Maryland
dmanocha@umd.edu